

SR001-VR1

RICO Citation Verification Report

Manuscript	RICO: Relationally-Induced Coherence Organization in Transformer Inference
Manuscript ID	SR001
Manuscript Version	v6.3, August 8, 2026
Report ID	SR001-VR1
Report Version	1.2
CVP Protocol	SF0037 v1.4
Verifier	Claude Opus 4.6 (claude-opus-4-6), Anthropic
Verification Date	August 8, 2026
Verification Tier	Tier 1 (Single-Platform Verified) with PAR persistence records; see Section 6
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Report Version Note

This is a substantive revision, not a reissue. Report versions v1.0 and v1.1 verified RICO v3.8 and v5.7 respectively. Those versions were previously described as permanently locked to the manuscript version they verified. That lock has been lifted by PCP decision of 2026-08-08, on the ground that the lock governs which manuscript version a report verifies, not whether the report's findings are correct, and v1.1 contained an error that could not be corrected without re-issuing.

The error: v1.1 recorded C2 (Olsson et al., 2022) as verified with no fabrication detected, and dispositioned it KEEP. The RICO reference at that time listed an author, "Rai, A.", who does not appear anywhere in the source, and gave DasSarma the wrong initial. The v1.1 record shows the verifier encountered the discrepancy and produced an unparseable note about it rather than assigning the AMBIGUOUS rating that CVP Section 5 Part C requires when support cannot be confidently determined. The certification statement issued on that basis was therefore inaccurate as to C2.

This version re-runs the full protocol against RICO v6.3, in which the reference has been corrected. The failure is documented rather than quietly repaired, because a demonstration artifact for a verification protocol is more useful to a practitioner when it shows the protocol's real failure mode than when it shows only a clean pass.

1. Live Browsing Access Confirmation

Per SF0037 v1.4 Section 4.1. Live access confirmed prior to verification. Full text was retrieved for every citation; no citation was assessed from abstract or metadata alone.

URL Opened	Content Confirmed	Status
arxiv.org/abs/2005.14165	Brown et al. (2020) landing page; 31-author byline and v4 submission history retrieved.	LIVE
arxiv.org/pdf/2005.14165	75-page full text extracted; few-shot definition located at p.6.	LIVE
transformer-circuits.pub/2022/in-context-learning-and-induction-heads/index.html	Full HTML retrieved (124,410 characters of body text); byline and induction-head definition confirmed.	LIVE
aclanthology.org/2024.tacl-1.9.pdf	17-page TACL full text; lost-in-the-middle finding located at p.1.	LIVE
arxiv.org/pdf/2404.06654	27-page RULER full text; context-length degradation	LIVE

	results located at p.1.	
arxiv.org/pdf/2309.17453	21-page ICLR 2024 full text; attention-sink mechanism located at p.1.	LIVE
doi.org/10.5281/zenodo.18289391	Concept DOI resolved to record 19155138 (CRD v1.6); PDF openly downloadable.	LIVE

2. Citation Inventory

Per SF0037 v1.4 Section 4.2. Total citations in RICO v6.3: N = 6.

ID	Reference	Cited In	Access Class
C1	Brown et al. (2020). Language models are few-shot learners. arXiv:2005.14165.	5.3	FREE-FULLTEXT
C2	Olsson et al. (2022). In-context learning and induction heads. Transformer Circuits Thread.	5.3, 7.2	FREE-FULLTEXT
C3	Liu et al. (2024). Lost in the middle. TACL, 12, 157-173.	1.2	FREE-FULLTEXT
C4	Hsieh et al. (2024). RULER. arXiv:2404.06654.	1.2	FREE-FULLTEXT
C5	Xiao et al. (2023). Attention sinks. arXiv:2309.17453.	7.3	FREE-FULLTEXT
C6	Gantz, T. W. (2026). Context representation drift. SF0039. Zenodo.	References only	FREE-FULLTEXT

3. Claim Map

Per SF0037 v1.4 Section 4.4. Claim text is verbatim from RICO v6.3.

Cite	Claim ID	Claim Text (verbatim)
C1, C2	CLAIM-01	in-context learning, where models adapt to patterns from examples provided within a prompt (Brown et al., 2020; Olsson et al., 2022)
C2	CLAIM-02	Olsson et al. (2022) describe induction heads -- circuit components that allow transformer models to continue patterns observed earlier in the context window
C3	CLAIM-03	Liu et al. (2024) demonstrated that models frequently underutilize information positioned in the middle of long contexts, retrieving content near the boundaries of a context more reliably than content buried in the middle
C4	CLAIM-04	the limits of effective context utilization across different architectures (Hsieh et al., 2024)
C5	CLAIM-05	Research on attention sink phenomena demonstrates that attention does not distribute evenly across context tokens (Xiao et al., 2023). Certain tokens accumulate disproportionate attention weight regardless of their semantic relevance.
C6	NONE	No inline claim in RICO v6.3 depends on this citation. It appears in the reference list only. See Section 4, C6, and Section 7 item 1.

4. Individual Citation Verification Records

C1 — Brown et al. (2020)

Reference	Brown, T. B., Mann, B., Ryder, N., Subbiah, M., Kaplan, J., Dhariwal, P., & Amodei, D. (2020). Language models are few-shot learners. arXiv:2005.14165.
Canonical ID	arXiv:2005.14165 DOI 10.48550/arXiv.2005.14165
Access Class	FREE-FULLTEXT

Retrieved At	https://arxiv.org/abs/2005.14165 and https://arxiv.org/pdf/2005.14165
Retrieval date	2026-08-08
Claims Supported	CLAIM-01

Evidence anchor — CLAIM-01

Pointer: p.6 (Section 2.1, Figure 2.1 discussion)

Excerpt: "few-shot works by giving K examples of context and completion"

Why this supports the claim: Defines few-shot operation as in-prompt example provision at inference time, which is the adaptation-from-in-prompt-examples the RICO claim attributes to the source.

Support rating: FULL SUPPORT

Bibliographic note: the source carries 31 authors. The RICO reference lists the first six followed by the final author, Amodei, with no ellipsis, which presents a truncated list as though complete. All seven named authors are correct and correctly initialled, and no name is fabricated; this is a style defect, not an integrity defect. APA would use the first nineteen authors, an ellipsis, then the final author. Recommendation recorded in Section 7.

Disposition: KEEP (bibliographic style note recorded)

C2 — Olsson et al. (2022)

Reference	Olsson, C., Elhage, N., Nanda, N., Joseph, N., DasSarma, N., Henighan, T., Mann, B., & Ndousse, K. (2022). In-context learning and induction heads. Transformer Circuits Thread.
Canonical ID	transformer-circuits.pub (first-party publication; no DOI issued)
Access Class	FREE-FULLTEXT
Retrieved At	https://transformer-circuits.pub/2022/in-context-learning-and-induction-heads/index.html
Retrieval date	2026-08-08
Claims Supported	CLAIM-01, CLAIM-02

Evidence anchor — CLAIM-01

Pointer: Introduction

Excerpt: "induction heads might constitute the mechanism for the actual majority of all in-context learning"

Why this supports the claim: Establishes the source as an in-context learning reference, which is the role the joint citation assigns it alongside Brown et al.

Support rating: FULL SUPPORT

Evidence anchor — CLAIM-02

Pointer: Section: induction head definition

Excerpt: "induction heads in our models are implemented by a circuit of two attention heads"

Why this supports the claim: The source defines induction heads as a two-head circuit that locates tokens preceded by the present token and continues the earlier pattern, which is precisely the pattern-continuation circuit the claim describes.

Support rating: FULL SUPPORT

AUTHOR LIST CORRECTION, superseding report v1.1. The reference verified in v1.1 read "...Mann, B., DasSarma, A., Ndousse, K., & Rai, A." Two defects were present and both were passed.

First: the source byline gives Nova DasSarma. The initial A. was wrong; it is N.

Second: no author named Rai appears in the source. An exhaustive search of the retrieved full text returns exactly one occurrence of the string "Rai", as a substring of "Rebecca Raible" in the contributions list, and zero occurrences of Rai as a standalone surname anywhere in the byline, the BibTeX block, or the acknowledgements. The name was fabricated.

Report v1.1 recorded that it had encountered this discrepancy and wrote that "Rai A. listed as 'Kamal Ndousse' in full list", a statement that does not resolve to any coherent finding, then concluded no fabrication detected and dispositioned KEEP. Under SF0037 Section 5 Part C, a verifier that cannot confidently determine the position must assign AMBIGUOUS, which is treated as failure. The correct v1.1 disposition was REVISE CLAIM.

RICO v6.3 corrects the reference to the source byline order: Olsson, Elhage, Nanda, Joseph, DasSarma (N.), Henighan, Mann, Ndousse. All eight names verified present and correctly initialled against the retrieved byline. Henighan, previously omitted, sits between DasSarma and Mann in the source and is restored.

Disposition: KEEP (manuscript corrected in v6.3; v1.1 disposition superseded)

C3 — Liu et al. (2024)

Reference	Liu, N. F., Lin, K., Hewitt, J., Paranjape, A., Bevilacqua, M., Petroni, F., & Liang, P. (2024). Lost in the middle: How language models use long contexts. Transactions of the Association for Computational Linguistics, 12, 157-173.
Canonical ID	ACL Anthology 2024.tacl-1.9 DOI 10.1162/tacl_a_00638
Access Class	FREE-FULLTEXT
Retrieved At	https://aclanthology.org/2024.tacl-1.9.pdf
Retrieval date	2026-08-08
Claims Supported	CLAIM-03

Evidence anchor — CLAIM-03

Pointer: p.1 (Abstract and Introduction; finding restated in Section 1)

Excerpt: "performance is often highest when relevant information occurs at the beginning or end of the input context"

Why this supports the claim: The source states the boundary-advantage and middle-degradation pattern directly, which is the whole of the RICO claim.

Support rating: FULL SUPPORT

Bibliographic note cleared: report v1.1 recorded that RICO cited the arXiv preprint numbering (vol. 11, 553-569, 2023) rather than the final TACL record. RICO v6.3 uses the final journal citation (vol. 12, 157-173, 2024) and the ACL Anthology locator. The v1.1 recommendation was adopted and this note is closed.

Disposition: KEEP

C4 — Hsieh et al. (2024)

Reference	Hsieh, C. Y., Sun, S., Krizan, S., Acharya, S., Rekesh, D., Jia, F., Zhang, Y., & Ginsburg, B. (2024). RULER: What's the real context size of your long-context language models? arXiv:2404.06654.
Canonical ID	arXiv:2404.06654 DOI 10.48550/arXiv.2404.06654
Access Class	FREE-FULLTEXT
Retrieved At	https://arxiv.org/pdf/2404.06654
Retrieval date	2026-08-08
Claims Supported	CLAIM-04

Evidence anchor — CLAIM-04

Pointer: p.1 (Abstract)

Excerpt: "almost all models exhibit large performance drops as the context length increases"

Why this supports the claim: Across the seventeen long-context models RULER evaluates, the source reports that only half sustain satisfactory performance at 32K, establishing limits on effective context utilisation across architectures as claimed.

Support rating: FULL SUPPORT**Disposition: KEEP****C5 — Xiao et al. (2023)**

Reference	Xiao, G., Tian, Y., Chen, B., Han, S., & Lewis, M. (2023). Efficient streaming language models with attention sinks. arXiv:2309.17453.
Canonical ID	arXiv:2309.17453 DOI 10.48550/arXiv.2309.17453 ICLR 2024
Access Class	FREE-FULLTEXT
Retrieved At	https://arxiv.org/pdf/2309.17453
Retrieval date	2026-08-08
Claims Supported	CLAIM-05

Evidence anchor — CLAIM-05

Pointer: p.1 (Abstract and Introduction)

Excerpt: "strong attention scores towards initial tokens as a sink even if they are not semantically important"

Why this supports the claim: Supports both halves of the claim: attention is unevenly distributed, and the tokens accumulating disproportionate weight do so independently of semantic relevance.

Support rating: FULL SUPPORT

Author list verified complete against the source byline: Guangxuan Xiao, Yuandong Tian, Beidi Chen, Song Han, Mike Lewis. Five authors, all present, correctly initialled, no truncation.

Disposition: KEEP**C6 — Gantz (2026), Context Representation Drift**

Reference	Gantz, T. W. (2026). Context representation drift. Synthience Institute. SF0039. DOI: 10.5281/zenodo.18289391.
Canonical ID	Concept DOI 10.5281/zenodo.18289391, resolving to version record 19155138 (v1.6)
Access Class	FREE-FULLTEXT
Retrieved At	https://doi.org/10.5281/zenodo.18289391
Retrieval date	2026-08-08
Claims Supported	NONE — see finding below

Evidence anchor — No claim mapped

Pointer: Not applicable

Excerpt: Not applicable; no manuscript claim depends on this citation.

Why this supports the claim: Support verification cannot be performed on a citation that supports no claim. Part C is not reached.

Support rating: NOT APPLICABLE (Part C not reached)

PVC status resolved. Report v1.1 assigned this citation UNABLE TO VERIFY because the Zenodo record did not resolve in any public index on 2026-03-05. It now resolves: the concept DOI redirects to version record 19155138, CRD v1.6, published under the Synthience Institute community with the full PDF openly downloadable (178.8 kB, md5 97f250409fedb3557089fb669f49f5dc). The v1.1 flag is cleared. That flag was correctly raised at the time and correctly did not exempt the author's own work.

NEW FINDING — floating citation, SF0037 v1.4 Section 4.4. The reference appears in RICO v6.3's reference list, and a full-document search finds no inline citation of Gantz, SF0039, CRD, or Context representation drift anywhere in the

body text. Section 4.4 states that citations not mapped to claims are invalid and must be removed or reattached with explicit claim linkage. Report v1.1 assessed this under CVP v1.3 and recorded the absence of an inline claim as the reason a PVC failure would not disqualify the manuscript; v1.4's Section 4.4 makes the same absence itself disqualifying.

This is not a fabrication or an accessibility problem. The cited work exists, is the author's own, is openly accessible, and is architecturally related to RICO. The defect is that the reference list asserts a dependency the manuscript does not have.

Disposition: REMOVE or REATTACH — PCP decision required. See Section 7 item 1.

5. Verification Summary

Metric	Result
Total citations inventoried (N)	6
Citations fully processed	6 / 6
Citations passing PVC (FREE-FULLTEXT)	6 (C1-C6)
Citations mapped to a manuscript claim	5 (C1-C5); C6 unmapped
Citations rated FULL SUPPORT	5 (C1-C5), across six claim mappings
Citations failing Section 4.4 claim-map requirement	1 (C6)
Citations removed or replaced	0 (one pending PCP decision)
Fabricated author names detected and corrected	1 (C2, corrected in manuscript v6.3)
Prior-report findings superseded	2 (C2 disposition; C6 PVC flag cleared)
Bibliographic style notes recorded	1 (C1 truncation without ellipsis)
Verification tier achieved	Tier 1 with PAR records; see note below

Tier note, stated plainly. SF0037 Section 7 sets Tier 2 (dual-platform verification) as the default for public release. This run achieved Tier 1: each citation was verified against one authoritative first-party source with full text opened and evidence anchors captured, and PAR persistence records (canonical identifiers plus retrieval URLs, and a file hash for C6) were recorded for all six. Independent second-repository confirmation was not performed for every citation. The honest tier is therefore Tier 1 with PAR, not Tier 2. Report v1.1 claimed Tier 2 on the basis of dual confirmation for two of six citations, which overstated the tier; that claim is superseded. Achieving Tier 2 requires a second independent repository check per citation and is recommended before this report is deposited.

6. Certification Statement

I certify that I executed SF0037 CVP v1.4 using live retrieval against public sources. Full text was opened for every citation; no support rating was assigned from an abstract, from metadata, or from model priors.

Total citations inventoried: N = 6

Citations fully processed: 6 / 6

Citations retained after PVC and support verification: 5 unconditional (C1-C5); 1 pending PCP disposition (C6)

Citations removed or replaced: 0

Verification tier achieved: Tier 1 (Single-Platform Verified) with PAR persistence records

Verification completed: August 8, 2026

Verifier: Claude Opus 4.6 (claude-opus-4-6), Anthropic | Prepared by: Thomas W. Gantz, Synthience Institute

7. Recommendations and Notes for Publication

1. C6 (Gantz 2026, CRD, SF0039) — action required. The citation satisfies PVC but fails the Section 4.4 claim-map requirement, because no body-text claim in RICO v6.3 depends on it. Two dispositions are available and the choice is a PCP decision. Remove it from the reference list, which costs nothing evidentially since no claim rests on it. Or reattach it by adding an inline citation at a point where RICO's argument genuinely draws on context representation drift; Section 1.2, which surveys long-context literature, is the natural site, and CRD is thematically adjacent to the Liu and Hsieh material already cited there. Reattachment is the stronger option if and only if the manuscript actually uses the source; adding an inline citation solely to legitimise a reference-list entry would invert the requirement.
2. C1 (Brown et al. 2020) — optional style correction. The seven-name list is a truncation of a 31-author byline presented without an ellipsis. All named authors are correct. If house style follows APA, the form is the first nineteen authors, an ellipsis, then the final author; the common short form is Brown, T. B., et al. (2020). No integrity issue and no urgency.
3. C2 (Olsson et al. 2022) — closed. The fabricated author and the incorrect initial are corrected in RICO v6.3 and verified against the source byline. No further manuscript action.
4. C3 (Liu et al. 2024) — closed. The final TACL citation replaced the preprint numbering per the v1.1 recommendation.
5. Tier upgrade before deposit. Section 5's tier note records that this run is Tier 1 with PAR. A second independent repository confirmation per citation would achieve the Tier 2 default that SF0037 Section 7 sets for public release.
6. Publication location. Report v1.1 was deposited as a standalone Zenodo record. Canon Published records that in hindsight it should have been included in the SF0037 record as a Supporting Artifact under a single DOI, and that future Supporting Artifacts will follow that pattern. This version supersedes v1.1 within the existing SR001-VR1 record, so no new DOI is minted and the structural anomaly is not repeated.
7. On the value of this report as a demonstration artifact. Report v1.1 was published to show what CVP output looks like. It now also shows what a verification failure looks like from the inside: a verifier that retrieved the correct source, encountered a name it could not place, wrote a fluent sentence that resolved nothing, and passed. Section 5 Part C already prescribes the correct behaviour, which is to rate AMBIGUOUS and treat it as failure. The gap was not in the protocol but in its execution, and the class of error is one an automated author-list comparison cannot commit, since a name either appears in the retrieved byline or it does not. Practitioners running CVP should treat any author-name discrepancy as AMBIGUOUS until resolved by direct string comparison against the retrieved source.